Lab 2 - GLM

Maths to GLM

```
#include <glm/glm.hpp>
```

$u = (0.6, 0.8, 0), \quad v = (-0.8, 0.6, 0)$

```
glm::vec3 u(0.6f, 0.8f, 0), v(-0.8f, 0.6f, 0);
```

$|u| = |v| = 1, \quad u \cdot v = 0$

```
glm::length(u); // 1
glm::dot(u, v); // 0
```

$w = u \times v = (0, 0, 1)$

```
glm::vec3 w;
w = glm::cross(u, v); // (0, 0, 1)
```

Maths to GLM

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	`glm::mat2 A(1, 3,` `2, 4);`
$a_{1,2} = 2 \quad (\text{row } 1, \text{column } 2)$	`A[1][0];`
$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$	`A * glm::vec2(1, 1); // (3, 7)`
$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$	`glm::transpose(A); // (1, 2, 3, 4)`
$\lvert A \rvert = 1 \cdot 4 - 2 \cdot 3 = -2$	`glm::determinant(A); // -2`
$A^{-1} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$	`glm::inverse(A); // (-2, 1.5, 1, -0.5)`

Exercise 1

Using the GLM Library, show that the cross product is distributive over addition using vectors:

$$\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$$

- Test this by passing random vectors to the method that will validate the equality
- Show intermediate results

Exercise 2

Using the GLM library, implement the following Rotation transformations as matrices. Apply them to the vectors below.

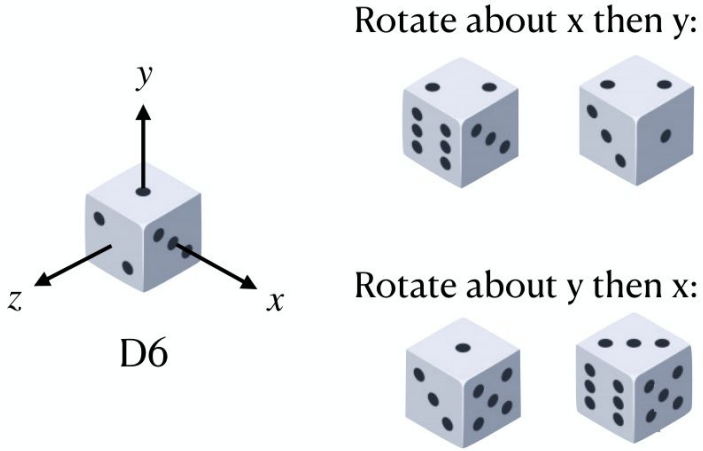
$$R_x = 45^o$$

$$R_y = 90^o$$

$$u = [1,1,1] \qquad v = [-1,-1,0] \quad w = [\ 1,0,0]$$

Exercise 3

Show that 3D Rotations are not commutative (unlike 2D Rotations).



- Test this by passing random vectors to the method that will validate the equality

Exercise 4

Consider a plane determined by the basis vectors a = [1, 2, 3] and b = [4, 5, 6]

- Analytically compute a vector orthogonal to this plane and verify your result.
  (Hint: both can be done via vector calculus operations)

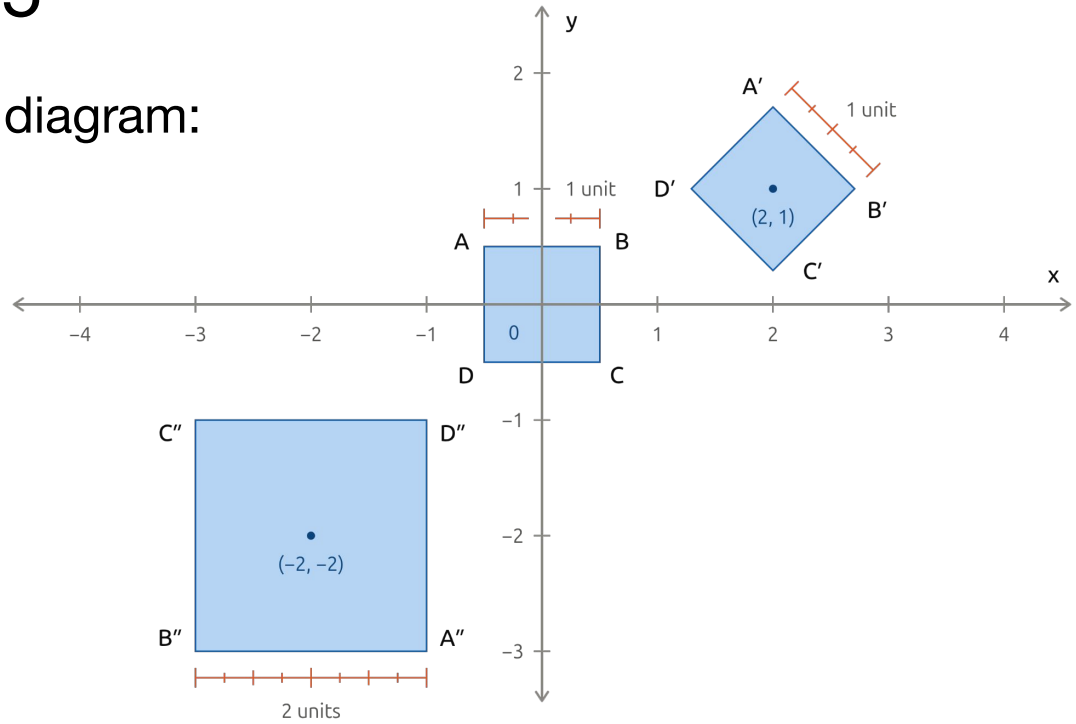
- Using the GLM library, implement two functions:
  1. orthogonalVector: receives two 3D vectors and returns a unit vector orthogonal to both
  2. areOrthogonal: receives three 3D vectors and returns whether the third is orthogonal to the first two

  (Assume that the input vectors are nonzero and not collinear)

- Test your implementation with the examples above and the 2D cartesian basis. What happens if you swap vectors in the first function?

Exercise 5

Consider the diagram:



- How would you encode the center unit square via a matrix?
- Regarding the center square, what sequence of transformations output the remaining squares?
- Implement both of your answers in GLM